25901584 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEZGYA00 - GENOME TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Continuous cell lines are often derived from: [U] / [CO1]

   a). Normal healthy tissues b). Cancerous or immortalized cells

   c). Plant sources d). Bacterial cultures

2. Which concentration of Ethanol is most commonly used for surface sterilization in cell culture? [K] / [CO1]

   a). 50% b). 70% c). 95% d). 100%

3. The best stage of mitosis for preparing a karyotype is: [K] / [CO2]

   a). Prophase b). Metaphase c). Anaphase d). Telophase

4. Which reagent is commonly used to precipitate DNA from aqueous solution? [K] / [CO2]

   a). Isopropanol or ethanol b). Phenol

c). Sodium dodecyl sulfate d). Proteinase K

5. What is the major advantage of CMA over Karyotyping? [An] / [CO2]

a). It is cheaper

b). It has a much higher resolution to detect micro-deletions/duplications

c). It can detect balanced translocations

d). It does not require DNA

6. Satellite DNA is mainly [U] / [CO2]

a). Protein-coding DNA b). Highly repetitive DNA

c). RNA coding DNA d). Single-copy DNA

7. In disease research, metabolomics can help identify: [U] / [CO3]

a). Mutated genes b). Altered metabolic pathways

c). Chromosome numbers d). DNA replication errors

8. Structural proteomics focuses on: [U] / [CO3]

a). DNA replication b). Determining 3D structure of proteins

c). RNA transcription d). Chromosome mapping

9. Which enzyme is used in RFLP analysis? [U] / [CO2]

a). DNA polymerase b). Restriction enzyme

c). Ligase d). Reverse transcriptase

10. Which analysis in MEGA assesses the reliability of the tree branches? [U] / [CO5]

a). BLAST b). Bootstrap

c). ORF finding d). Annotation

[1x10 = 10]

Part B

Short Answer Questions

Answer any **14** questions

Each question carries **2** marks

11. Explain why a cell culture laboratory must maintain positive air pressure relative to the surrounding hallways. [U] / [CO1]
12. How is colony formation assessed in cell culture? [U] / [CO1]
13. Explain the concept of passage number and its significance in cell culture maintenance. [U] / [CO1]
14. What is the basic purpose of CRISPR technology in genetics? [U] / [CO2]
15. Describe why SNPs are considered useful markers in genomics. [E] / [CO2]
16. Why is maintaining sterility especially important during tissue processing? [U] / [CO1]
17. Name the chemicals used for base-specific cleavage in Maxam–Gilbert sequencing. [K] / [CO3]
18. Define Quantitative Fluorescent PCR (QF-PCR). [K] / [CO2]
19. State two advantages of NGS over traditional sequencing methods. [U] / [CO3]
20. Mention various applications of VNTRs in medical genetics. [E] / [CO2]
21. Describe how transcriptomics helps in understanding disease mechanisms. [E] / [CO3]
22. Why is WES focused on exons rather than introns? [E] / [CO3]
23. Explain the difference between Global and Local sequence alignment. [U] / [CO5]
24. How does the "Minimum ORF Length" parameter help reduce false positives? [U] / [CO4]
25. What specific information is found in the 'FEATURES' table of a GenBank record? [U] / [CO4]
26. Explain the purpose of the Variant Effect Predictor (VEP) in Ensembl. [U] / [CO4]

[2x14 = 28]

Part C

Short Essay Questions

Answer any **8** questions

Each question carries **4** marks

27. Explain the major components of media (salts, glucose, amino acids, vitamins, serum, antibiotics) and their roles. [U] / [CO1]

28. Explain the step-by-step procedure for starting a primary cell culture from a biological specimen. [U] / [CO1]

29. Explain the function of Taq polymerase in PCR and its importance in the amplification process. [U] / [CO2]

30. Analyze the principles and applications of quantitative PCR (qPCR), explaining how it enables precise measurement of initial DNA or RNA quantities. [An] / [CO2]

31. Describe the role of next-generation sequencing in diagnostic gene panels and evaluate its effectiveness. [E] / [CO3]

32. Evaluate whether Targeted Exome Sequencing is suitable for prenatal or pediatric genetic testing. [E] / [CO3]

33. Evaluate the role of primer design in determining the success and specificity of Sanger sequencing. [E] / [CO3]

34. Discuss the scope and significance of Bioinformatics in modern genomic research. [U] / [CO4]

35. Detail the utility of literature databases like PubMed in genomics [U] / [CO4]

36. What is meant by "Genome Annotation"? Distinguish between structural and functional annotation [U] / [CO4]

[4x8 = 32]